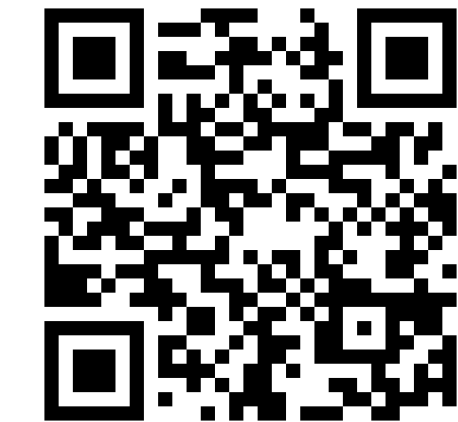


Agentic AI for Earth Science Software: a field report from building Worldscope

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An interactive Earth-system workbench in which AI agents are not chatbots bolted on, but first-class users of every capability.



See it run
Source & walkthrough video

WHAT IT DOES

Built in 4½ months. Every commit written by an AI agent.

An **open-source Earth-2 tool** for weather, climate, and Earth-observation data, built to lower the barrier to entry for people who think visually. It runs on any device but uses whatever NVIDIA hardware is present for data services and AI inference — with a single desktop **DGX Spark** it runs the vast majority of AI weather models locally. It complements **Earth-2 Command Center**, which remains the right tool for photorealistic visualization and Omniverse digital twins.

DATA, FILES & FORMATS

- Reads **NetCDF**, **GRIB2**, **Zarr**, **GeoTIFF** across many grids — **HEALPix rendered natively**, not interpolated to lat-lon first
- Opens remote files; moves whole files or single fields between machines
- Many online sources: weather, climate, satellite observation
- Active weather caching, so animation plays at speed
- Every display carries units and a legend

MODELS & LIVE EARTH

- Launches **Earth2Studio** runs from a GUI — forecast, diagnostic, assimilation
- Ensemble member, vertical level, and time are all selectable axes
- Live conditions with animated winds and currents, from several sources
- Live feeds: aircraft, ships, earthquakes, satellites
- Meteograms at a point; side-by-side compare through a swipe window

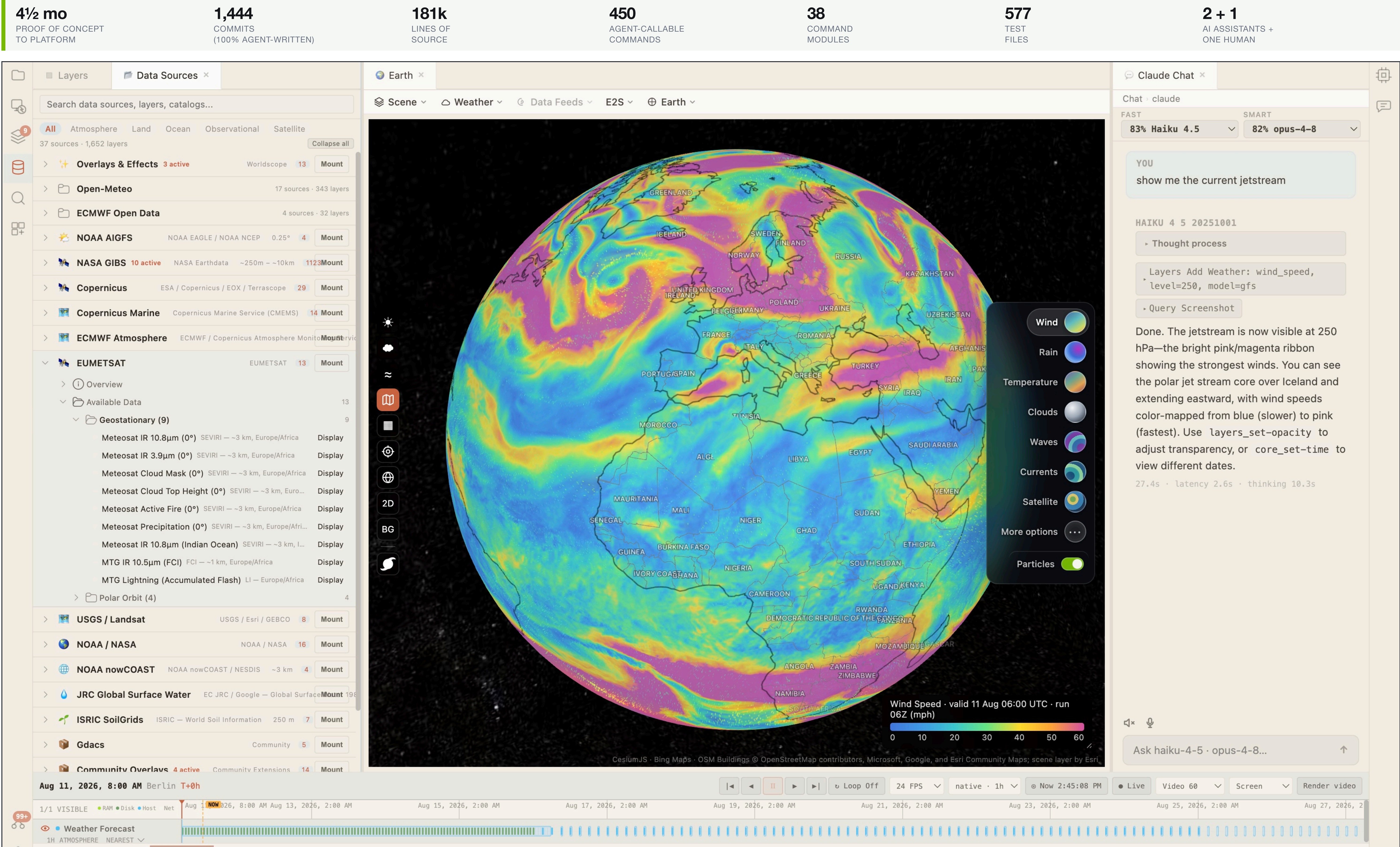
VIEWS & OUTPUT

- 2D and 3D at every scale, globe down to city blocks with buildings
- Sun, Moon, Mars and other bodies, for planetary missions
- Exports images, PDFs, and animations at chosen format and frame rate
- Multi-window, multi-tab shell modeled on VS Code; demo mode hides the UI
- iPhone and iPad touch support in development

AGENTS AS USERS

- Every capability sits behind an **MCP server** — the same 450 commands the UI itself calls, not a chat window bolted on
- An agent loads data, runs a model, compares results, and drives the same scene the human sees
- **Several agents can work one scene at once**
- AI-driven demos: a slideshow of features that stays fully interactive

NEXT weather radar · Jupyter notebooks · large ensembles and forecast tracks · vertical-profile analysis · Earth-observation workflows such as change detection · AI model chains running global forecast → regional impact → decision support



Worldscope. A global forecast field on the interactive globe, with the layer explorer at left and an agent session at right. The agent drives the same scene the human sees — same layers, same camera, same time cursor — through the Model Context Protocol, not a scripting sidecar.

One project is a data point. The pattern is the story.

I The work did not vanish. It moved.

For well-specified features, agents produced implementations roughly **a hundred times faster** than I could have written them. That collapsed the cost of implementation and pushed the effort into specification, integration, and verification — none of which got cheaper.

4 Tests helped. Tests were not enough.

Agent-written tests caught regressions, but **could not establish that a complex interactive workflow behaved correctly end to end**. Plausible, runnable and quietly wrong is the failure mode that matters most in scientific software — and the one a green test suite does not catch.

2 Vision became the bottleneck.

Holding a coherent whole mattered **more** as implementation got cheaper, not less. Left to themselves, agents converge on the generic: the ordinary layout, the common grid, the default colormap, the solution that already exists somewhere in the training data.

5 Parallelism cuts both ways.

Running agents in parallel accelerated implementation while multiplying conflicting assumptions and integration regressions. Meanwhile the tools themselves were in constant flux, each new assistant arriving with its own strengths, weaknesses, and personality quirks.

I served as architect, tester, and director; the agents wrote the code. What follows is what that felt like from the inside — and the questions it raises for everyone here, because none of them are specific to this project.

3 Verification was the hardest part.

Especially for interactive behavior, and partly for a structural reason: the agents observed the browser through **still screenshots** and had to infer what happened in between. Features routinely looked correct and did not work; fixing one silently broke another.

Where the effort went

Schematic — impressions from one project, not measurements.

WRITING IT MYSELF

Implementation	Spec	Integ.	Verify
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DIRECTING AGENTS

Specification	Integration	Verification
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■ Writing code ■ Saying what it must do ■ Making the parts cohere ■ Proving it is right

OPEN QUESTIONS FOR THIS COMMUNITY

Answers welcome. Arguments preferred.

I WHERE THE WORK GOES

- What happens when AI agents are treated not as chatbots, but as *full participants* in an Earth-science workflow?
- If writing code becomes a hundred times cheaper, which part of the scientific workflow becomes the bottleneck — and is any group staffed for it?
- Is the durable artifact still the source code, or the *specification* that regenerates it?
- What is scientific programming a skill *in*, once the programming is free?

IV PROVENANCE & REVIEW

- If a published result rests on 180,000 lines no human has read line by line, what exactly was peer-reviewed?
- What should journals and programme committees require — prompts, model versions, tool versions, transcripts, seeds?
- Is agent-generated code reproducible once the agent that wrote it has been deprecated?
- Who is the author of a scientific tool, and what counts as a citable software contribution now?

II VERIFICATION & PHYSICALITY

- How do you verify software you did not write, at the rate it is now written?
- Who catches code that runs, passes its tests, and is *physically* wrong — flipped signs, unit slips, staggered-grid offsets, silent regridding, mishandled missing values, the wrong CRS?
- Generation scales; checking does not. Is that asymmetry the real ceiling on agentic science?
- What is the software equivalent of a physicality check — conservation, invariance, dimensional consistency, known-answer tests?

V THE GENERIC ATTRACTOR

- Agents regress toward the mean of their training data. Does that quietly pull the field toward common grids, common formats, and mainstream methods?
- How much of scientific novelty lives precisely in the choices an agent would talk you out of?
- Does agentic development accelerate science, or accelerate its *consolidation*?
- Do we get more science — or the same science, faster?

III EVALUATION & BENCHMARKS

- We benchmark the models. Should we benchmark the *code that produces them* — and what would that benchmark measure?
- Agents perceive interactive systems as still frames and infer the motion in between. What does that imply for evaluating anything time-dependent, interactive, or visual?
- Do agent-built pipelines quietly alter our datasets — resampling, gap-filling, reprojecting — in ways our metrics never see?
- If an agent writes the model, the training loop, *and* the evaluation, what independence is left?

VI PEOPLE, ACCESS, COMMONS

- If early-career scientists never write the code, where does the judgment to review it come from — and what do we train them to be good at?
- Does this widen the gap between well-resourced and under-resourced groups, or erase it?
- When a bespoke tool costs a weekend, do we still build shared community software — or does the toolchain quietly fragment?
- Is scientific software still worth *maintaining*, or is it now disposable and regenerable?

As the cost of intelligence drops rapidly toward zero, the scarce resource turns out to be **artistic vision and judgment**.

If that is right —
are we selecting for it, teaching it,
or rewarding it?